Ego dissolution as a collapse of integration, not a surge of entropy: a whole-brain dynamic mean-field model of combined serotonergic and NMDA-antagonist action

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Abstract

The "entropic brain" hypothesis holds that psychedelic and other ego-dissolving states correspond to an increase in the entropy or signal diversity of spontaneous brain activity. Motivated by a first-person report of a combined serotonergic–dissociative state (LSD/DMT together with nitrous oxide, N_2O), we asked a sharper question: when a 5-HT_{2A} agonist and an NMDA antagonist act *together*, is the resulting state dominated by a rise in local signal complexity, by a collapse of long-range functional integration, or by both? We address this in a validated whole-brain dynamic mean-field (DMF) model (reduced Wong–Wang, Deco et al. 2014) implemented in The Virtual Brain, on empirical human structural connectomes, with regional excitation/inhibition balanced by analytic Feedback Inhibition Control (FIC). The two drug classes are represented by *separable, biophysically motivated* parameters: 5-HT_{2A} agonism as a density-weighted increase in excitatory response gain, and sub-anaesthetic NMDA antagonism as interneuron-preferential disinhibition (reduced excitatory→inhibitory NMDA conductance) plus reduced long-range NMDA coupling. Crucially, because both perturbations shift mean firing rate — and both Lempel–Ziv complexity (LZc) and functional connectivity (FC) co-vary with rate — we **rate-match** every condition back to the ∼3 Hz physiological set-point before reading out the metrics, isolating mechanism from a trivial rate confound. At matched rate, the combination produces a **robust, dose-dependent, topology-general collapse of functional integration** (mean ΔFC $= -56\% \pm 12\%$ SD, negative in 6/6 noise seeds and reproduced on a second 192-region connectome) together with a **small but statistically reproducible increase in signal complexity** (ΔLZc $= +0.9\% \pm 0.5\%$ SD, positive in 6/6 seeds, $t \approx 4$). These multi-seed values are the canonical results; single-run figures quoted in the development log are individual noise realisations. The 5-HT_{2A} entropy increase (+5.4%) is largely cancelled by an N_2O-driven hypersynchrony (−4.6%), so the net complexity change is marginal while the integration collapse is large. We conclude that, in this model, the combined ego-dissolving state is **dissociation-dominated rather than entropy-dominated**: the brain loses its capacity to integrate, not its local richness. We discuss this as a refinement of the entropic-brain framing for combined serotonergic–dissociative states, and report the analyses transparently including a working-point artefact we explicitly reject.

Keywords: ego dissolution, entropic brain, dynamic mean-field model, The Virtual Brain, feedback inhibition control, 5-HT_{2A}, NMDA antagonist, nitrous oxide, functional integration, Lempel–Ziv complexity.

1. Introduction

Classic serotonergic psychedelics (LSD, psilocybin, DMT) acutely increase the diversity or entropy of spontaneous cortical activity, a finding that underpins the *entropic brain* hypothesis [Carhart-Harris et al., 2014]. Magnetoencephalography studies report increased Lempel–Ziv signal complexity not only for psychedelics but also for the dissociative NMDA antagonist ketamine [Schartner et al., 2017], suggesting that "elevated complexity" may be a common signature of several ego-dissolving states. In parallel, a separate line of work links the psychedelic ego-dissolution experience specifically to changes in *global functional connectivity* and network integration [Tagliazucchi et al., 2016].

These two readouts — local signal complexity and global integration — are conceptually distinct, and they need not move together. A state could be locally more complex yet globally less integrated, or vice versa. This distinction becomes acute when *two* pharmacologically distinct ego-dissolving agents are combined. The present work is motivated by a first-person report of exactly such a combination: a serotonergic psychedelic (LSD/DMT) taken together with the dissociative anaesthetic nitrous oxide (N_2O), producing an unusually complete dissolution of self. The phenomenological question — "does the self dissolve because the brain becomes maximally complex, or because it can no longer integrate?" — has a precise computational counterpart that we set out to test.

We adopt a whole-brain dynamic mean-field (DMF) approach [Deco et al., 2014], which represents each brain region by coupled excitatory and inhibitory neural populations on an empirical structural connectome, and which has an established method for representing serotonergic neuromodulation through receptor-density-weighted gain [Deco et al., 2018]. The two drug classes map onto *separable* model parameters, which is what makes the combination tractable: 5-HT_{2A} agonism as excitatory gain, NMDA antagonism as interneuron-preferential disinhibition plus reduced long-range coupling.

A central methodological point organises the study. Both perturbations shift the model's mean firing rate, and both LZc and FC depend on firing rate; therefore any naïve comparison risks reporting a firing-rate artefact rather than a mechanism. We control for this by re-balancing inhibition (Feedback Inhibition Control) so that every condition sits at the same ∼3 Hz set-point before metrics are read — a within-model "rate-matched" design.

2. Methods

2.1 Whole-brain dynamic mean-field model

We use the reduced Wong–Wang excitatory–inhibitory DMF [Deco et al., 2014] as implemented in The Virtual Brain (tvb-library 2.10) [Sanz Leon et al., 2013]. Each region k carries an excitatory (E) and inhibitory (I) population with NMDA-gated synaptic variables S_E, S_I:

- $I_{E,k} = W_E \cdot I_0 + w_p \cdot J_N \cdot S_{E,k} + G \cdot J_N \cdot \sum_j C_{kj} S_{E,j} - J_{i,k} \cdot S_{I,k}$
- $I_{I,k} = W_{EI} \cdot J_N \cdot S_{E,k} - S_{I,k} + W_I \cdot I_0$
- firing rates $r = H(I)$ via the standard sigmoidal transfer functions H_E, H_I;
- $dS_E/dt = -S_E/\tau_E + (1 - S_E) \cdot \gamma \cdot H_E + \text{noise}$; $\quad dS_I/dt = -S_I/\tau_I + \gamma_I \cdot H_I + \text{noise}$.

Parameters follow Deco et al. (2014): a_E=310, b_E=125, d_E=0.16; a_I=615, b_I=177, d_I=0.087; γ=0.000641, γ_I=0.001; τ_E=100 ms, τ_I=10 ms; W_E=1, W_I=0.7, I_0=0.382 nA; w_p=1.4; J_N=0.15 nA. Integration: stochastic Heun, dt=1 ms, additive noise σ=1e−4. State initialised at the low-activity (≈3 Hz) fixed point to avoid the bistable high-activity branch.

2.2 Connectome and working point

Structural connectivity is the TVB default empirical human connectome — the `connectivity_76.zip` dataset shipped with *tvb-data* and loaded via `Connectivity.from_file()` (a 76-region Hagmann-type DTI parcellation; regional centroids in MNI-like coordinates) — with distance-dependent conduction delays (tract lengths up to ~153 mm, speed 3 mm/ms). Weights are normalised to unit maximum. Because the normalised row-sums are ≈13, the effective global coupling is large; the stable low-activity working point for this connectome is therefore $G \approx 0.06$, just below the transition to runaway high-activity (edge between G=0.06 stable and G=0.08 unstable under analytic FIC). A 192-region connectome is used for replication (§3.5).

2.3 Feedback Inhibition Control (analytic + polish)

To hold every region near a physiological 3.06 Hz, regional inhibitory weights J_i are set by **analytic FIC**: at the set-point all S_E equal the known value 0.164, the target excitatory input current is $I_E^* = (b_E - 8)/a_E = 0.377$ nA (the −0.026 nA balance point), and the inhibitory fixed point S_I^* is obtained by bisection, giving a closed-form per-region

$$J_{i,k} = (W_E{\cdot}I_0 + w_p{\cdot}J_N{\cdot}0.164 + G{\cdot}J_N{\cdot}0.164{\cdot}\kappa_k - I_E^*)/S_I^*,$$

with κ_k the node strength. A short stochastic refinement ($\leq$ 25 iterations) polishes residual error. This procedure clamps the baseline to $\langle S_E \rangle = 0.164$, $r_E = 3.07$ Hz (max regional deviation 0.012), which the earlier hand-rolled and proportional-only controllers could not achieve on the heterogeneous empirical topology.

2.4 Pharmacological perturbations (separable, biophysical)

- **5-HT$_{2A}$ agonism (LSD/DMT):** increase the excitatory response gain a_E by a receptor-density-weighted factor $(1 + s{\cdot}\rho_k)$, with ρ a 0–1 density map and s the gain magnitude (default s=0.15) [after Deco et al., 2018].
- **NMDA antagonism (N_2O), faithful model:** sub-anaesthetic NMDA block is interneuron-preferential, producing disinhibition. We subclass the model with a separate excitatory→inhibitory NMDA conductance $J_{N,EI}$ and reduce it (default −40%), modelling disinhibition; we additionally reduce the long-range coupling G (default −30%), modelling reduced NMDA-mediated integration.
- For comparison, a *crude* N_2O model (uniform reduction of J_N) was also evaluated.

The receptor-density map ρ is a synthetic monotone gradient with randomised regional assignment; its specific spatial pattern is varied across seeds (§2.6) to test robustness, pending substitution by an empirical 5-HT$_{2A}$ PET atlas.

2.5 Readouts and the rate-matching control

From the excitatory gating time series S_E (sampled every 10 ms) we compute: (i) **Lempel–Ziv complexity** (LZ76, normalised) per region on the median-binarised signal, averaged over regions; (ii) **functional integration** as the mean upper-triangular Pearson correlation (FC) across regions. A haemodynamic (Balloon–Windkessel) BOLD monitor was also evaluated for FC.

Rate-matching: because both LZc and FC co-vary with firing rate, FIC is re-tuned *per condition* so that every condition (baseline, 5-HT$_{2A}$, N_2O, combination) sits at 3 Hz. Reported effects are therefore at matched rate, isolating mechanism from rate.

2.6 Robustness protocol

We assessed: (1) multi-seed reproducibility (n=6 independent noise realisations, each with a different randomised 5-HT_{2A} map); (2) dose–response (combined dose scaled 0.25 → 1.0); (3) a second, larger connectome (192 regions); (4) a working-point (G) sweep for the FC-magnitude regime.

3. Results

3.1 The validated baseline

Analytic FIC clamps the empirical-connectome baseline to $r_E = 3.07$ Hz (max regional deviation 0.012), a stable physiological resting point. We note that a hand-implemented Euler DMF with proportional-only FIC failed to reach this regime (collapsing to 0 Hz or diverging to 10^2–10^3 Hz across coupling values), which motivated both the validated integrator and the analytic FIC; the key practical bug was the coupling scale ($G \approx 0.06$, not ≈ 2, for a connectome with row-sums ≈ 13).

3.2 Single agents reproduce expected signatures

At matched rate, the 5-HT_{2A} condition reproduces the canonical psychedelic signature — increased signal complexity (ΔLZc = +5.4% ± 0.5, positive in 6/6 seeds) — consistent with the entropic-brain literature [Carhart-Harris et al., 2014; Schartner et al., 2017]. The faithful (disinhibitory) N_2O condition, by contrast, *decreases* complexity (ΔLZc = −4.6% ± 0.3, negative in 6/6 seeds): in this model disinhibition drives hypersynchrony rather than entropy. Both single agents reduce functional integration (5-HT_{2A} −46%, N_2O −18%).

3.3 The combination is dissociation-dominated

At matched rate, the combination produces:

Condition (rate-matched ∼3 Hz)	**ΔLZc**	**ΔFC integration**
5-HT_{2A} (LSD/DMT)	+5.4% ± 0.5 (6/6)	−46% ± 14 (6/6)
N_2O (faithful, disinhibitory)	−4.6% ± 0.3 (0/6)	−18% ± 7 (6/6)
Combination	**+0.9% ± 0.5** (6/6)	**−56% ± 12** (6/6)

The combination's defining feature is the **large, robust collapse of integration** (−56%), with only a **small net complexity increase** (+0.9%): the 5-HT_{2A} entropy boost is largely cancelled by N_2O-induced hypersynchrony (+5.4% − 4.6% ≈ +0.8%). The small ΔLZc is nonetheless statistically reliable (6/6 seeds positive; mean/standard-error ≈ 4).

3.4 Dose–response

Scaling the combined dose deepens the integration collapse monotonically: ΔFC = −25%, −41%, −52%, −59% at doses 0.25, 0.50, 0.75, 1.00 (rate-matched), with ΔLZc steady at +1.0…+1.6%. A monotone dose–response is the signature of a genuine mechanism rather than a single-point coincidence.

3.5 Replication on a second connectome

On an independent 192-region empirical connectome, the combination reproduces the pattern: ΔFC = −49%, ΔLZc = +1.2% (two seeds). The effect is not specific to one parcellation.

3.6 Working point, FC magnitude, and empirical receptor map

With analytic FIC the model remains stable up to at least G=0.10 ($r_E \approx 3.6$ Hz), and mean baseline FC increases monotonically with G (gating-FC $0.0043 \to 0.0137$ over $G = 0.04 \to 0.10$). A working-point calibration sweeping (G, noise) with FIC re-tuning raised the haemodynamic (BOLD) baseline FC from ≈0.02 (G=0.06) to ≈0.043 (G=0.10); raising the noise further destabilised the 3 Hz set-point (runaway to 6–12 Hz). At the higher-FC point (G=0.10) the combination's integration collapse was confirmed (BOLD ΔFC = −63%). Absolute FC stays below empirical resting magnitudes — an inherent property of a rate-clamped mean-field resting state — but the verdict, being a *relative* change, is independent of this.

Repeating the **full multi-seed analysis** (n = 6) with the **empirical Beliveau Cimbi-36 5-HT$_{2A}$ map** in place of the synthetic gradient gave combination ΔLZc = +1.5% ± 0.3 and Δintegration = −61% ± 19 (`results_realmap_multiseed.csv`; per-seed rows in `results_realmap_multiseed_perseed.csv`) — statistically indistinguishable from the synthetic-map result (+0.9% ± 0.5; −56%±12; `results_multiseed.csv`, per-seed in `results_multiseed_perseed.csv`). The verdict is therefore independent of whether the receptor map is synthetic or empirical. (A single empirical-map run had given −75% — `results_realmap.csv` — an individual noise realisation within the −35…−84% per-seed spread, illustrating why the multi-seed mean, not any single run, is the reported quantity; §5, limitation 1.)

3.7 A rejected artefact (reported for transparency)

Before rate-matching, the combination at the critical working point drove mean firing to ∼24 Hz and the metrics flagged the naïve hypothesis as "supported." We reject this as a working-point artefact: at 24 Hz the model is far from its FIC-calibrated regime, and the apparent effect reflects the hot dynamical state, not the intended mechanism. Only the rate-matched analysis is reported as the result.

4. Discussion

In this model, the combined serotonergic–dissociative state is **dissociation-dominated, not entropy-dominated**. The robust, dose-dependent, topology-general finding is a collapse of long-range functional integration; the change in local signal complexity, while statistically reliable, is small because the 5-HT$_{2A}$-driven entropy increase and an N_2O-driven hypersynchrony nearly cancel.

This asymmetry — a large integration collapse (−56%) against a near-null complexity change (+1%) — is worth stating plainly, because it **partly contradicts the expectation that motivated the study**. The originating first-person report, and the entropic-brain intuition more generally, would predict a *large* rise in signal complexity; the model instead returns essentially none, while the dissociative axis dominates. A model that disconfirms the naïve prediction of the very experience that prompted it is, by that token, not a confirmation-seeking exercise: the $-56\%/+1\%$ asymmetry is the result's strongest internal check against motivated reasoning.

This refines, rather than contradicts, the entropic-brain framing. For a *single* serotonergic psychedelic, the model reproduces the expected complexity increase. It is the *combination* with an NMDA antagonist — specifically, the disinhibitory hypersynchrony the antagonist contributes — that cancels the entropy signature while compounding the loss of integration. The phenomenological reading is that, under the combination, the self dissolves because the brain loses its capacity to *integrate* distributed activity into a unified model, not because local activity becomes maximally disordered. This is consistent with accounts linking ego dissolution to global connectivity change [Tagliazucchi et al., 2016] and complements complexity-centred accounts by separating the two axes.

An incidental modelling result is that *faithful* (disinhibitory) NMDA antagonism reduced complexity in this DMF, contrary to a naïve "disinhibition → more entropy" expectation. Whether this reflects a genuine prediction or a limitation of the mean-field representation of interneuron-specific block is an open question (see Limitations).

5. Limitations

This is a mechanistic, hypothesis-generating model, and several limitations bound its claims:

1. **Receptor map.** We incorporated the empirical Beliveau et al. (2017) 5-HT_{2A} atlas — the [^{11}C]Cimbi-36 PET map (the same dataset used by Deco et al. 2018), obtained from the *netneurolab/hansen_receptors* repository — by sampling the volume at each region centroid. The empirical map, run through the full multi-seed protocol ($n = 6$), gave the same answer as the synthetic gradient (combination Δintegration $-61\% \pm 19$ vs $-56\% \pm 12$; ΔLZc $+1.5\% \pm 0.3$ vs $+0.9\% \pm 0.5$ — statistically indistinguishable), confirming map-independence with real receptor data. Caveat: the TVB Hagmann connectome's centroids are only approximately in MNI152 space, so the region-level sampling is imperfect ($\sim\frac{1}{3}$ of the 76 centroids fell outside cortical tissue and were median-filled). This approximation is acceptable because the multi-seed analysis independently established that the verdict does not depend on the receptor map's spatial detail; a connectome supplied with a matched parcellation would sharpen the mapping.
2. **Absolute FC magnitude.** A working-point calibration (sweeping G and noise with FIC retuning) roughly doubled baseline BOLD FC ($\approx$0.02 $\rightarrow$ $\approx$0.043 at G=0.10) but did not reach empirical resting magnitudes, and higher noise destabilised the set-point. The integration collapse was confirmed at this higher-FC point (-63%). The residual gap reflects the rate-clamped mean-field resting state and bears only on absolute magnitude, not on the relative verdict.
3. **Single model family.** Results are within the reduced Wong–Wang DMF; replication in an independent model family (e.g., a spiking or a different neural-mass model) would strengthen generality.
4. **Phenomenological pharmacology.** The drugs are represented by a small number of effective parameters. Interneuron-preferential NMDA block is modelled as a reduced E$\rightarrow$I conductance plus reduced coupling; richer receptor-level and laminar detail is absent.
5. **No direct empirical comparison.** The model is validated against published single-drug *signatures* (sign of the LZc change), not fitted to combined-state neuroimaging data, which to our knowledge does not yet exist for this specific combination.

These limitations are stated to bound the result, not to diminish it: the central finding (a robust, dose-dependent, topology-general integration collapse with a small reproducible complexity increase) survives the robustness checks performed.

6. Conclusion

A validated whole-brain dynamic mean-field model, with separable serotonergic and NMDA-antagonist mechanisms and a rate-matched design that removes a firing-rate confound, indicates that the combined psychedelic–nitrous-oxide ego-dissolving state is characterised primarily by a **collapse of functional integration** rather than by a surge of signal entropy. The result is robust across noise realisations and receptor maps, scales with dose, and generalises to a second connectome. It reframes the phenomenology of this combined state as a loss of integration — the unbinding of a distributed self-model — with local signal complexity largely preserved.

Data and code availability

All simulation and analysis code, result tables, figures, and the full development log are openly available at github.com/Biniruprojects/ego-dissolution-dmf — model drivers, analytic FIC, the rate-matching control, and the multi-seed/dose/second-connectome/G-sweep robustness scripts (`whole_brain_tvb*.py`), result tables (`results_*.csv`), figures (`FIG*.png`), and the step-by-step development log (`FINDINGS.md`). Built on tvb-library 2.10 (Python 3.13) and reproducible with fixed seeds. The empirical 5-HT_{2A} map is the Beliveau et al. (2017) [^{11}C]Cimbi-36 atlas obtained via netneurolab/hansen_receptors.

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